

MUHAMMAD MAHEEM

AI Developer & Machine Learning Engineer

mirza.muhammad.maheem@gmail.com +92 300-8714141 LinkedIn GitHub Portfolio Lahore, Pakistan

ABOUT ME

21-year-old AI Developer and Machine Learning Engineer focused on building intelligent systems that solve real-world problems. Experienced across the entire development lifecycle—from data preprocessing and model training to deploying production-ready applications with robust authentication and responsive UIs. Specialized in RAG systems, Computer Vision pipelines, and Reinforcement Learning.

PROFESSIONAL EXPERIENCE

AI/Full-Stack Developer Intern

Punjab Skill Development Authority (PSDA)

Mar 2026 – Present · Paid · Remote

- Building a production SaaS application from Figma designs: website, admin dashboard, backend (PostgreSQL), and frontend (Next.js + Tailwind CSS).
- Developed role-based authentication, responsive UI, and RESTful API integrations for real-world users.
- Collaborating with cross-functional teams to deliver scalable, maintainable solutions on tight deadlines.

Frontend Developer

Artificial Automation Business Solutions (AABS)

Sep 2025 – Nov 2025 · Part-time · Remote

- Developed responsive UI components for a full-stack SaaS Management Dashboard using Next.js 15, React 19, and TypeScript.
- Implemented secure user authentication flows with NextAuth, analytics visualizations, and subscription management interfaces.
- Collaborated asynchronously with backend teams to integrate RESTful APIs and ensure seamless data flow in production environments.

FLAGSHIP PROJECTS

Quality Evaluation PDF Chatbot – Production-Ready RAG Pipeline

- Automated RAG evaluation using RAGAS metrics (Faithfulness, Context Relevance, Answer Relevance) with a quality gate (≥ 0.80).
- CLI workflow for batch and single-question evaluation, golden dataset, and JSON/summary reporting for CI/CD integration.
- Stack: Python, FastAPI, ChromaDB, Groq/Llama 3, LangGraph, LangChain, Prompt Engineering.

RAG-Based PDF Chatbot (Production-Ready)

- Developed a secure end-to-end RAG system using FastAPI, Next.js 14+, and Groq (Llama 3 70B).
- Built a robust document pipeline: text chunking → Sentence Transformers embeddings → ChromaDB vector storage.
- Implemented JWT authentication, isolated user workspaces, and efficient PDF retrieval for context-aware responses.

Hadith Chatbot

- Indexed 39,038 authentic Hadiths using FAISS vector search and LangChain for semantic retrieval.
- Developed a Flask-based backend capable of answering complex queries based on religious texts with high accuracy.
- Optimized for fast inference and precise cross-referencing across large-scale textual datasets.

Tic-Tac-Toe AI via Deep Q-Network (DQN)

- Trained a Reinforcement Learning agent in PyTorch over 100,000 self-play episodes.
- Implemented experience replay and epsilon-greedy exploration to achieve a near-optimal game strategy (win/draw rate $>98\%$).
- Developed model persistence scripts for seamless loading and real-time interaction via a CLI interface.

Additional Technical Projects

Face Recognition Attendance with Anti-Spoofing: Dlib/OpenCV liveness detection, Streamlit GUI, and automated attendance logging.

Value Notes: Chrome extension with React 18, TypeScript, Tailwind CSS, and real-time Firebase sync.

Bank Management System: Functional banking app using Python OOP and Streamlit GUI.

TECHNICAL SKILLS

AI & Machine Learning

Python · PyTorch · Scikit-learn · OpenCV · FAISS · RAG Systems · LangChain · LangGraph · ChromaDB · Reinforcement Learning · Prompt Engineering

Web Development

Next.js 14/15 · React 18/19 · FastAPI · TypeScript · JavaScript · Tailwind CSS · PostgreSQL · Firebase · Streamlit · REST APIs

Tools & Practices

Git/GitHub · OOP · Linux · Claude Code · Copilot CLI · Data Structures · Problem-Solving · Technical Communication · Self-Directed Learning

EDUCATION

Superior University, Lahore | 2024–2028 | BSc Artificial Intelligence (4th Sem, CGPA: 3.58/4.0) | Focus: ML, DL, NLP, CV, DSA, Linear Algebra, Stats